

Lie Theory: Assignment 4

钟皓宇(12533556)

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Problem 1

Prove that $\Phi^\vee$ is a root system in E , whose Weyl group is naturally isomorphic to $\mathcal{W}$; show also that $\langle \alpha^\vee, \beta^\vee \rangle = \langle \beta, \alpha \rangle$, and draw a picture of $\Phi^\vee$ in the cases A_1, A_2, B_2, G_2 .

Solution: R1 holds for $\Phi^\vee$. Since Φ is finite then $\Phi^\vee$ is also finite. Hence Φ does not contain zero implies that $\Phi^\vee$ also does not contain zero. Φ and $\Phi^\vee$ can be linearly represented for each other, it implies that $\Phi^\vee$ spans E . R2 holds for $\Phi^\vee$ for the reason that the dual operator just change the norm but not change the angle. To see that we have calculation:

$$\cos \theta_{\alpha^\vee, \beta^\vee} = \frac{(\alpha^\vee, \beta^\vee)}{\|\alpha^\vee\| \cdot \|\beta^\vee\|} = \frac{4(\alpha, \beta)}{\|\alpha\|^2 \cdot \|\beta\|^2} \cdot \frac{\|\alpha\| \cdot \|\beta\|}{4} = \frac{(\alpha, \beta)}{\|\alpha\| \cdot \|\beta\|} = \cos \theta_{\alpha, \beta},$$

for arbitrary elements $\alpha, \beta \in \Phi$. Hence, we find that $(-\alpha)^\vee = -\alpha^\vee$. Therefore, R2 holds. For R3, we see that

$$\sigma_{\alpha^\vee}(\beta^\vee) = \beta^\vee - \frac{2(\beta^\vee, \alpha^\vee)}{\|\alpha^\vee\|^2} \alpha^\vee$$

Since $\alpha^\vee / \|\alpha^\vee\| = \alpha / \|\alpha\|$ we obtain that

$$\sigma_{\alpha^\vee}(\beta^\vee) = \sigma_\alpha(\beta^\vee) = \frac{2\beta}{\|\beta\|^2} - \frac{4(\beta, \alpha)}{\|\beta\|^2 \cdot \|\alpha\|^2} \alpha = \frac{2}{\|\beta\|^2} \sigma_\alpha(\beta).$$

Because σ_α is a reflection, it does not change the norm. Moreover, σ_α leaves Φ invariant. Then we find that

$$\sigma_{\alpha^\vee}(\beta^\vee) = (\sigma_\alpha(\beta))^\vee \in \Phi^\vee.$$

It shows that $\sigma_{\alpha^\vee}$ leaves $\Phi^\vee$ invariant. R4 is also true. Because

$$\langle \beta^\vee, \alpha^\vee \rangle = (\beta^\vee, (\alpha^\vee)^\vee) = (\beta^\vee, \alpha) = \langle \alpha, \beta \rangle \in \mathbb{Z}.$$

Therefore, we obtain that $\Phi^\vee$ is also a root system. In the previous discussion, we can see that $\sigma_\alpha = \sigma_{\alpha^\vee}$. Therefore, the Weyl groups of Φ and $\Phi^\vee$ are the same subgroup of $\text{GL}(E)$ since they share the same generators. The picture of $\Phi^\vee$ as follow. ■

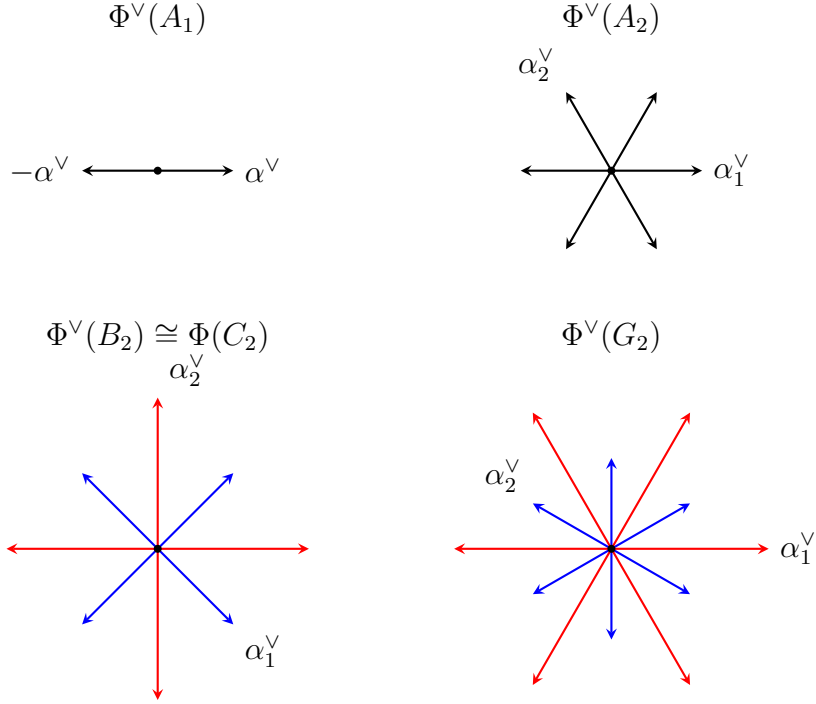


Figure 1: The dual root systems $\Phi^\vee$ for A_1, A_2, B_2, G_2 .

Problem 2

Let $\Phi^\vee$ be the dual system of Φ , $\Delta^\vee = \{\alpha^\vee \mid \alpha \in \Delta\}$. Prove that $\Delta^\vee$ is a base of $\Phi^\vee$. [Compare Weyl chambers of Φ and $\Phi^\vee$.]

Solution: Since the dual operator does not change the hyperplane. Explicitly, we say $P_\alpha = P_{\alpha^\vee}$. It implies that Φ and $\Phi^\vee$ share the same Weyl chambers. Suppose that $\Delta = \{\alpha_1, \dots, \alpha_n\}$, then it define a Weyl chamber

$$C(\Delta) = \{x \in E \mid (x, \alpha_i) > 0 \text{ for } i = 1, \dots, n\}.$$

Equivalently, we obtain that

$$C(\Delta) = \{x \in E \mid (x, \alpha_i^\vee) > 0 \text{ for } i = 1, \dots, n\}.$$

According to our textbook, $C(\Delta)$ as the Weyl chamber of $\Phi^\vee$ can be uniquely determined by roots $\{\beta_1^\vee, \dots, \beta_n^\vee\}$ such that

$$C(\Delta) = \{x \in E \mid (x, \beta_i^\vee) > 0 \text{ for } i = 1, \dots, n\},$$

and $\{\beta_i\}$ form a base. It implies that $\Delta^\vee = \{\beta_1, \dots, \beta_n\}$ and thus $\Delta^\vee$ is a base of $\Phi^\vee$. ■

Problem 3

Given $\Delta = \{\alpha_1, \dots, \alpha_l\}$ in Φ , let $\lambda = \sum_{i=1}^l k_i \alpha_i$ ($k_i \in \mathbf{Z}$, all $k_i \geq 0$ or all $k_i \leq 0$). Prove that either λ is a multiple (possibly 0) of a root, or else there exists $\sigma \in \mathcal{W}$ such that $\sigma\lambda = \sum_{i=1}^l k'_i \alpha_i$, with some $k'_i > 0$ and some $k'_i < 0$. [Sketch of proof: If λ is not a

multiple of any root, then the hyperplane P_λ orthogonal to λ is not included in $\bigcup_{\alpha \in \Phi} P_\alpha$. Take $\mu \in P_\lambda - \bigcup_{\alpha \in \Phi} P_\alpha$. Then find $\sigma \in \mathcal{W}$ for which all $(\alpha_i, \sigma\mu) > 0$. It follows that $0 = (\lambda, \mu) = (\sigma\lambda, \sigma\mu) = \sum k'_i(\alpha_i, \sigma\mu)$.]

Solution: We use the notation of hint and follow its discussion. Let $C(\Delta)$ denote the fundamental Weyl chamber determined by Δ . Since μ is a regular vector, equivalently, we say there exists a Weyl chamber containing μ . According to our textbook, the Weyl group acts transitively on Weyl chambers, then there exists $\sigma \in \mathcal{W}$ such that $\sigma(\mu) \in C(\Delta)$. It implies that $(\alpha_i, \sigma(\mu)) > 0$ for all $i = 1, \dots, l$. Because we have

$$0 = (\lambda, \mu) = (\sigma\lambda, \sigma\mu) = \sum k'_i(\alpha_i, \sigma\mu).$$

The positivity of $(\alpha_i, \sigma\mu)$ implies the set $\{k'_i\}$ must contains negative and positive integers. Otherwise, it must equal to $\{0\}$ meaning that $\lambda = 0$. It is impossible, since in this case, $\lambda = 0 \cdot \alpha_1$ is a multiple of a root. ■

Problem 4

Prove that if α, β are roots, $\beta \neq \pm\alpha$, k is a real number and $\beta + k\alpha$ is a root, then k is an integer.

Solution: Let γ denote $\beta + k\alpha$. First, we consider that

$$\langle \gamma, \alpha \rangle = 2k + \langle \beta, \alpha \rangle \in \mathbb{Z}.$$

It requires that $k \in (1/2) \cdot \mathbb{Z}$. Hence, we consider

$$\langle \gamma, \beta \rangle = 2 + k\langle \alpha, \beta \rangle \in \mathbb{Z}.$$

Suppose that $k = m/2$ is not an integer, then we must have $\langle \alpha, \beta \rangle \in 2\mathbb{Z}$. Hence, since $-k\alpha = \beta - \gamma$ we consider

$$-k\langle \alpha, \gamma \rangle = \langle \beta, \gamma \rangle - 2 \in \mathbb{Z}$$

implies that $\langle \alpha, \gamma \rangle \in 2\mathbb{Z}$. Similarly, we consider $k\alpha = \gamma - \beta$ and

$$k\langle \alpha, \beta \rangle = \langle \gamma, \beta \rangle - 2 \in \mathbb{Z}.$$

It implies that $\langle \alpha, \beta \rangle \in 2\mathbb{Z}$. Since we have $\langle \alpha, \beta \rangle \in \{0, \pm 1, \pm 2, \pm 3\}$, then $\langle \alpha, \beta \rangle$ must be 0 or ± 2 . If $\langle \alpha, \beta \rangle = 0$, then we have $\langle \gamma, \beta \rangle = 2$. It means that $\langle \beta, \gamma \rangle = 1$ and $(\gamma, \gamma) = 2(\beta, \beta)$. Moreover, we have

$$(\gamma, \gamma) = (\beta + k\alpha, \beta + k\alpha) = (\beta, \beta) + k^2(\alpha, \alpha) + 2k(\alpha, \beta)$$

It implies that $(\beta, \beta) = k^2(\alpha, \alpha)$. Since we have

$$\langle \alpha, \gamma \rangle = \frac{2(\alpha, \beta + k\alpha)}{(\gamma, \gamma)} = \frac{2k(\alpha, \alpha)}{2(\beta, \beta)} = \frac{k(\alpha, \alpha)}{k^2(\alpha, \alpha)} = \frac{1}{k} = \frac{2}{m} \in 2\mathbb{Z}$$

we obtain that $m = \pm 1$. If $m = 1$, then we have

$$(\alpha, \alpha) = 4(\beta, \beta) = 2(\gamma, \gamma)$$

and $s_\gamma(\beta) = -\frac{1}{2}\alpha \in \Phi$ which is the multiple of α . It is impossible. If $m = -1$, by the same reason, we have $s_\gamma(\beta) = \frac{1}{2}\alpha \in \Phi$, which is also impossible. Suppose that $\langle \alpha, \beta \rangle = \pm 2$ then we have

$$(\alpha, \alpha) = 2(\beta, \beta) \quad \langle \beta, \alpha \rangle = \pm 1 \quad (\alpha, \beta) = \pm(\beta, \beta)$$

Hence, we have

$$(\gamma, \gamma) = (1 \pm 2k + 2k^2)(\beta, \beta).$$

Moreover, we obtain that

$$\langle \alpha, \gamma \rangle = \frac{\pm 2 + 2m}{1 \pm m + m^2/2} \in 2\mathbb{Z}$$

. Rewrite this formula, we obtain that

$$\frac{2(m \pm 1)}{(m \pm 1)^2 + 1} \in \mathbb{Z}.$$

Since m is odd, then $x = m \pm 1$ is even. The upper formula is equal to say

$$\frac{2x}{x^2 + 1} \in \mathbb{Z}$$

If $|x| \geq 2$, then we find that $|\frac{2x}{x^2+1}| < 1$, it implies that $x = 0$ and thus $m = \mp 1$. Then we find that

$$(\gamma, \gamma) = (1 - 1 + 1/2)(\beta, \beta) = \frac{1}{2}(\beta, \beta).$$

It implies that $(\alpha, \alpha) = 2(\beta, \beta) = 4(\gamma, \gamma)$ which is impossible. Therefore, k must be an integer. ■